Three-Dimensional Numerical Solutions

Three dimensional numerical solutions based on surface singularity distributions are similar, in principle, to methods presented for the two-dimensional case. From the theoretical aspect, only the wake and the trailing-edge conditions (three-dimensional Kutta condition) will require some additional attention. The most difficult aspect in three dimensions, though, is the modeling of the geometry, especially when arbitrary geometry capability is sought.

In the first part of this chapter the geometry (of the wing) is kept relatively simple and the aerodynamics of a thin lifting surface is modeled. In principle, this simple method has all the elements of the more complex panel methods and is capable of modeling the effect of wing planform shape on the fluid dynamic loads. In addition, the examples that are being presented require only limited programming effort and, therefore, are suitable for classroom instruction. Furthermore, the introduction in class of the numerical lifting-line model (Section 12.1), next to Prandtl's lifting-line model of Section 8.1, provides additional insight and a clear explanation of the spanwise integral equation.

In the second part of this chapter the principles of panel codes capable of solving the flow over bodies with arbitrary three-dimensional geometry will be presented. Over the years many such methods were developed and improved, but recent trends indicate an increased use of the approach that is based on the combination of surface source and doublet distributions with the inner potential boundary condition (for closed bodies). Consequently, only this approach will be presented through a brief description of one low-order and one high-order panel method.

In terms of classroom instruction it is recommended at this phase that use be made of one of the commercially available panel codes and that students be trained first to use the pre- and post-processor. This graphic pre-processor generates the surface grid (panel model) that is used to define the input to the computer program. The post-processor is usually a graphic utility that allows for a rapid analysis of the three-dimesional results by using extensive graphic representations. It is believed that after studying and preparing examples with the lifting surface code in this chapter (Section 12.3) students can safely proceed to use a larger panel code since at this phase they must have a deep understanding of the formulation and the construction of these codes.

12.1 Lifting-Line Solution by Horseshoe Elements

As a first example, consider the numerical solution of the lifting-line problem of Section 8.1. This can help us to understand the assumptions and limitations of the single vortex line method, which in this numerical form can be extended easily to include effects of wing sweep, dihedral, or even side slip. For simplicity (and in the spirit of Prandtl's model) only one chordwise vortex is used here but the method can easily be extended to include more chordwise vortices. The small-disturbance assumption of Chapter 8 still holds for this case, and a thin lifting wing with large aspect ratio ($\mathcal{R} > 4$) is assumed. This problem is

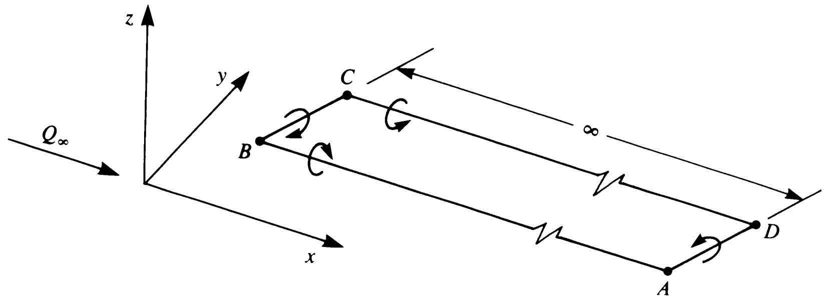

Figure 12.1 "Horseshoe" model of a lifting wing.

stated in terms of a vortex distribution in Section 4.5 and the following horseshoe model can be considered as the simplest approach to its solution.

In regard to solving Laplace's equation, the vortex line is a solution of this equation and the only boundary condition that needs to be satisfied is the zero normal flow across the thin wing's solid surface:

$$\nabla(\Phi + \Phi_\infty) \cdot \mathbf{n} = 0 \tag{12.1}$$

In the classical case of Prandtl's lifting-line model, the wing is placed on the x–y plane and then this boundary condition requires that the sum of the normal velocity component induced by the wing's bound vortices w_b, by the wake w_i and by the free-stream velocity $\mathbf{Q}_\infty$, will be zero (see also Eq. (8.2a)):

$$w_b + w_i + Q_\infty \alpha = 0 \tag{12.2}$$

Based on the proposed horseshoe element and on the above boundary condition, let us construct a numerical solution, following the six-step procedure of Chapter 9.

a. *Choice of Singularity Element*

To solve this problem the horseshoe element shown in Fig. 12.1 is selected. This element consists of a straight bound vortex segment (BC in Fig. 12.1) that models the lifting properties and of two semi-infinite trailing vortex lines that model the wake. The segment BC does not necessarily have to be parallel to the y axis, but at the element tips the vortex is shed into the flow where it must be parallel to the streamlines so that no force will act on the trailing vortices. In order not to violate the Helmholtz condition, these vortex elements are viewed as the near portions of vortex rings whose starting vortices extend far back, so that the effect of this segment (AD in Fig. 12.1) is negligible. The requirement that the far wake must be parallel to the free stream poses some modeling difficulties (which were not raised at all when constructing the classical lifting-line model). This is illustrated in Fig. 12.2a, which shows that the trailing wake has to be bent near the trailing edge to meet this free wake condition. Another possibility is shown in Fig. 12.2b, where the simple horseshoe vortex is kept, but the trailing segments are not shed at the trailing edge. Of course the very small angle of attack assumption (as in the case of the lifting-line model) allows the placing of the wake on the x–y plane of the body coordinate system as shown in Fig. 12.3. Since in this section the numerical solution of the lifting-line model is attempted, we shall adopt the model shown in Fig. 12.3, which assumes small angles of attack. However, the method can easily be modified to use the wake model as presented in Fig. 12.2a, and an even more detailed model will be presented in Section 12.3.

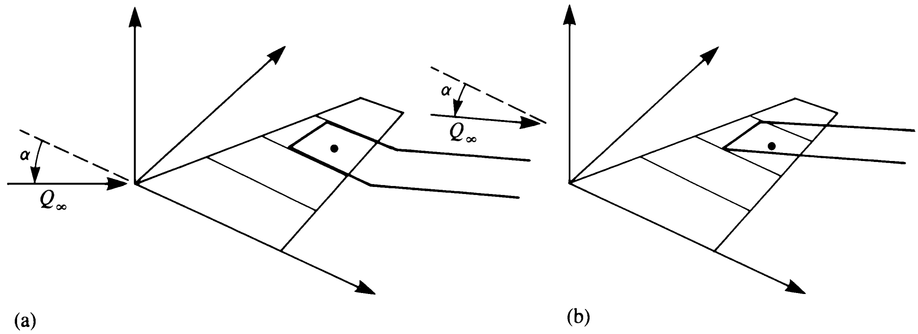

Figure 12.2 Difficulties of meeting the "wake parallel to local velocity" condition by a single horseshoe vortex representation.

The method by which the thin wing planform is divided into elements is shown in Fig. 12.3 and a typical spanwise element is shown in Fig. 12.4. Here, based on the results of the lumped-vortex model, the bound vortex is placed at the panel quarter chord line and the collocation point is at the center of the panel's three-quarter chord line. The strength of the vortex Γ is assumed to be constant for the horseshoe element and a positive circulation is defined as shown in the figure. Since this element is based on the lumped-vortex model, which includes the two-dimensional Kutta condition, it is assumed that this three-dimensional model accounts (in an approximate way) for the Kutta condition:

$$\gamma_{T.E.} = 0 \tag{12.3}$$

where the subscript *T.E.* stands for trailing edge. The velocity induced by such an element at an arbitrary point $P(x, y, z)$, shown in Fig. 12.4, can be computed by three applications

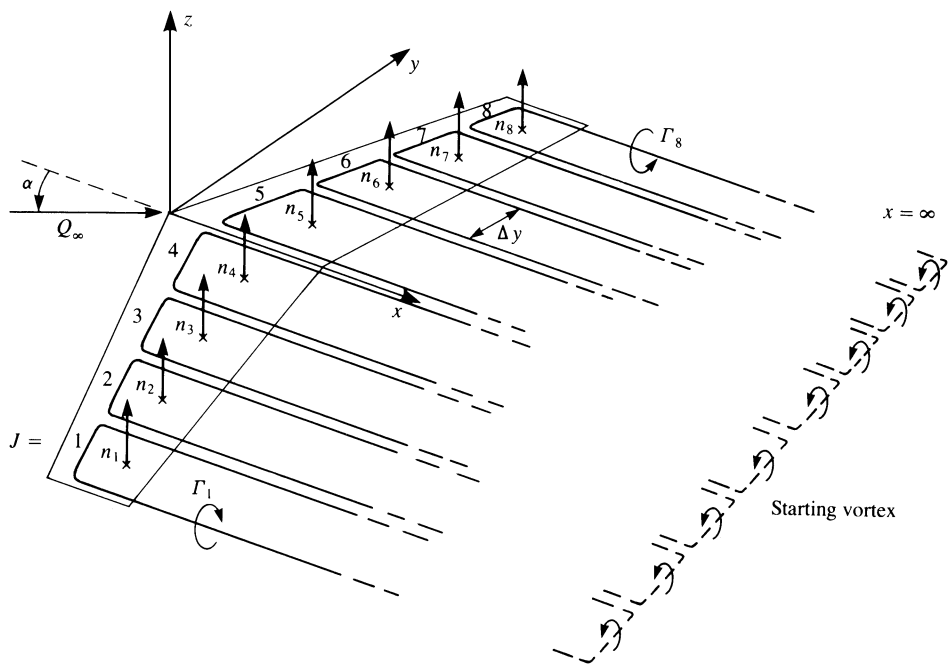

Figure 12.3 Horseshoe vortex lattice model for solving the lifting-line problem.

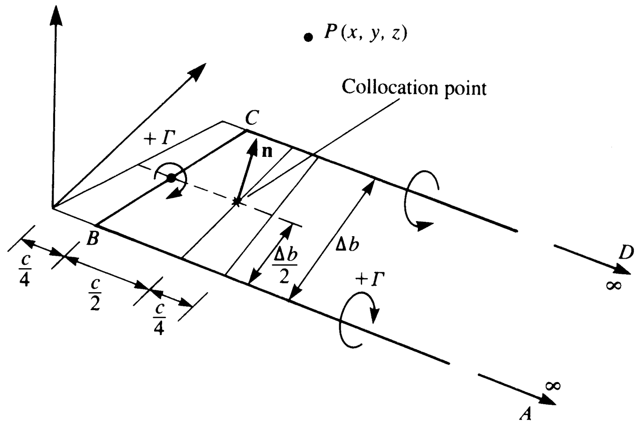

Figure 12.4 A spanwise horseshoe vortex element.

of the vortex line routine VORTXL (Eq. (10.116)) of Section 10.4.5:

$$\begin{aligned}(u_1, v_1, w_1) &= \mathrm{VORTXL}(x, y, z, x_A, y_A, z_A, x_B, y_B, z_B, \Gamma)\\ (u_2, v_2, w_2) &= \mathrm{VORTXL}(x, y, z, x_B, y_B, z_B, x_C, y_C, z_C, \Gamma)\\ (u_3, v_3, w_3) &= \mathrm{VORTXL}(x, y, z, x_C, y_C, z_C, x_D, y_D, z_D, \Gamma)\end{aligned} \tag{12.4}$$

At this point, let us follow the small-disturbance lifting-line approach and assume

$$y_A = y_B, \qquad y_C = y_D, \qquad \text{and} \qquad x_A = x_D \to \infty$$

Of course ∞ means that the influence of the vortex line beyond x_A or x_D is negligible, which from the practical point of view means at least twenty wing spans behind the wing. It is possible, at this point, to align the wake with the free stream by adjusting the points at $x = \infty$ (e.g., $z_A = x_A \sin\alpha$ and $z_D = x_D \sin\alpha$). It is also possible to use the model of Fig. 12.2a, which requires breaking the two trailing vortex segments into two segments each and computing their induced velocity in a similar manner.

The velocity induced by the three vortex segments is then

$$(u, v, w) = (u_1, v_1, w_1) + (u_2, v_2, w_2) + (u_3, v_3, w_3) \tag{12.4a}$$

It is convenient to include these computations (Eqs. (12.4) and (12.4*a*)) in a subroutine such that

$$(u, v, w) = \mathrm{HSHOE}(x, y, z, x_A, y_A, z_A, x_B, y_B, z_B, x_C, y_C, z_C, x_D, y_D, z_D, \Gamma)$$

It is recommended at this point that the trailing vortex wake-induced downwash $(u, v, w)^*$ be separated from the velocity induced by the bound vortex segments and saved. This information is needed for the induced-drag computations and if done at this phase will only slightly increase the computational effort. The influence of the trailing segment is obtained by simply omitting the (u_2, v_2, w_2) part from Eq. (12.4*a*):

$$(u, v, w)^* = (u_1, v_1, w_1) + (u_3, v_3, w_3) \tag{12.5}$$

So, at this point it is assumed that $(u, v, w)^*$ is automatically obtained as a byproduct of subroutine HSHOE.

b. *Discretization and Grid Generation*

At this phase the wing is divided into N spanwise elements as shown by Fig. 12.3 (with the panel side edge assumed to be parallel to the x axis). For this example the span is divided equally into $N = 8$ panels, and the spanwise counter j will have values between 1 and N. Also, geometrical information such as the panel area S_j, normal vector $\mathbf{n}_j$, and the coordinates of the collocation points (x_i, y_i, z_i) are calculated at this phase. For example, if the panel is approximated by a flat plate then the normal $\mathbf{n}_j$ is a function of the local angle α_j as defined in Fig. 11.3 or Fig. 11.17:

$$\mathbf{n}_j = (\sin\alpha_j, \cos\alpha_j) \tag{12.6}$$

c. *Influence Coefficients*

To fulfill the boundary conditions, Eq. (12.2) is specified at each of the collocation points (see Fig. 12.3). The velocity induced by the horseshoe vortex element no. 1 at collocation point no. 1 (hence the use of the index 1,1) can be computed by using the HSHOE routine developed before:

$$\begin{aligned}(u, v, w)_{11} = \text{HSHOE}(&x_1, y_1, z_1, x_{A1}, y_{A1}, z_{A1}, x_{B1}, y_{B1}, z_{B1},\\ &x_{C1}, y_{C1}, z_{C1}, x_{D1}, y_{D1}, z_{D1}, \Gamma = 1.0)\end{aligned}$$

Note that $\Gamma = 1$ is used to evaluate the influence coefficient due to a unit strength vortex. Similarly, the velocity induced by the second vortex at the first collocation point will be

$$\begin{aligned}(u, v, w)_{12} = \text{HSHOE}(&x_1, y_1, z_1, x_{A2}, y_{A2}, z_{A2}, x_{B2}, y_{B2}, z_{B2},\\ &x_{C2}, y_{C2}, z_{C2}, x_{D2}, y_{D2}, z_{D2}, \Gamma = 1.0)\end{aligned}$$

The no normal flow across the wing boundary condition (Eq. (12.2)), at this point, can be rewritten for the first collocation point as

$$\begin{aligned}[(u, v, w)_{11}\Gamma_1 &+ (u, v, w)_{12}\Gamma_2 + (u, v, w)_{13}\Gamma_3 + \cdots\\ &+ (u, v, w)_{1N}\Gamma_N + (U_\infty, V_\infty, W_\infty)] \cdot \mathbf{n}_1 = 0\end{aligned}$$

and the strengths of the vortices Γ_j are not known at this phase. Establishing the same procedure for each of the collocation points results in the discretized form of the boundary condition:

$$\begin{aligned}
a_{11}\Gamma_1 + a_{12}\Gamma_2 + a_{13}\Gamma_3 + \cdots + a_{1N}\Gamma_N &= -\mathbf{Q}_\infty \cdot \mathbf{n}_1\\
a_{21}\Gamma_1 + a_{22}\Gamma_2 + a_{23}\Gamma_3 + \cdots + a_{2N}\Gamma_N &= -\mathbf{Q}_\infty \cdot \mathbf{n}_2\\
a_{31}\Gamma_1 + a_{32}\Gamma_2 + a_{33}\Gamma_3 + \cdots + a_{3N}\Gamma_N &= -\mathbf{Q}_\infty \cdot \mathbf{n}_3\\
&\vdots\\
a_{N1}\Gamma_1 + a_{N2}\Gamma_2 + a_{N3}\Gamma_3 + \cdots + a_{NN}\Gamma_N &= -\mathbf{Q}_\infty \cdot \mathbf{n}_N
\end{aligned}$$

where the influence coefficients are defined as

$$a_{ij} \equiv (u, v, w)_{ij} \cdot \mathbf{n}_i \tag{12.7}$$

The normal velocity components of the free-stream flow $\mathbf{Q}_\infty \cdot \mathbf{n}_i$ are known and moved to the right-hand side of the equation:

$$\text{RHS}_i \equiv -(U_\infty, V_\infty, W_\infty) \cdot \mathbf{n}_i \tag{12.8}$$

We now have a set of N linear algebraic equations with N unknown Γ_j that can be solved by standard matrix solution techniques.

For example, for the case of a planar wing with constant angle of attack α, this results in the following set of equations:

$$\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1N} \\ a_{21} & a_{22} & \dots & a_{2N} \\ a_{31} & a_{32} & \dots & a_{3N} \\ \vdots & & \ddots & \vdots \\ a_{N1} & a_{N2} & \dots & a_{NN} \end{pmatrix} \begin{pmatrix} \Gamma_1 \\ \Gamma_2 \\ \Gamma_3 \\ \vdots \\ \Gamma_N \end{pmatrix} = -Q_\infty \sin\alpha \begin{pmatrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}$$

In practice it is recommended that two DO loops be used to automate the computation of the a_{ij} coefficients. The first will scan the collocation points, and the inner loop will scan the vortex elements for each collocation point:

DO 1 $i = 1, N$ (collocation point loop)
$\text{RHS}_i = -\mathbf{Q}_\infty \cdot \mathbf{n}_i$
DO 1 $j = 1, N$ (vortex element loop)

$$(u, v, w)_{ij} = \text{HSHOE}(x_i, y_i, z_i, x_{Aj}, y_{Aj}, z_{Aj}, x_{Bj}, y_{Bj}, z_{Bj}, x_{Cj}, y_{Cj}, z_{Cj}, x_{Dj}, y_{Dj}, z_{Dj}, \Gamma = 1.0)$$

$$a_{ij} = (u, v, w)_{ij} \cdot \mathbf{n}_i$$

$$b_{ij} = (u, v, w)^*_{ij} \cdot \mathbf{n}_i$$

1 END

Here b_{ij} is the normal component of the wake-induced downwash that will be used for the induced-drag computations and $(u, v, w)^*_{ij}$ is given by Eq. (12.5).

d. *Establish RHS Vector*

The right-hand side vector, Eq. (12.8), is actually the normal component of the free stream, which can be computed within the outer DO loop of the influence coefficient computations (as shown above). However, if one plans to upgrade the code by including unsteady effects or the simulation of normal "transpiration" flows, then this calculation should be done separately.

e. *Solve Linear Set of Equations*

The solution of the above described problem can be obtained by standard matrix methods. Furthermore, since the influence of such an element on itself is the largest, the matrix will have a dominant diagonal, and the solution is stable.

f. *Secondary Computations: Pressures, Loads, Velocities, Etc.*

The solution of the above set of equations results in the vector $(\Gamma_1, \Gamma_2, \dots, \Gamma_N)$. The lift of each bound vortex segment is obtained by using the Kutta–Joukowski theorem:

$$\Delta L_j = \rho Q_\infty \Gamma_j \Delta y_j \tag{12.9}$$

where Δy_j is the panel bound vortex projection normal to the free stream (see Fig. 12.4 where the panel width $\Delta b = \Delta y$). The induced-drag computation is somwhat more complex. Following the lifting-line results of Eq. (8.20a), we have

$$\Delta D_j = -\rho w_{\text{ind}_j} \Gamma_j \Delta y_j \tag{12.10}$$

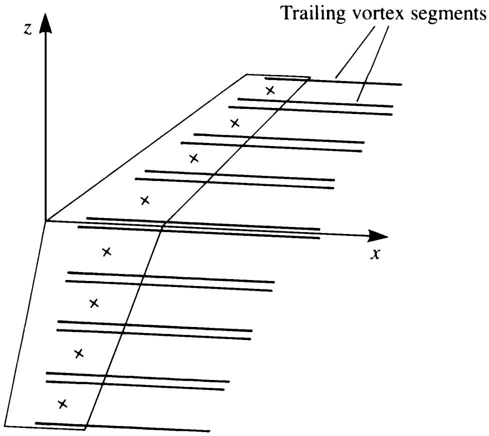

Figure 12.5 Array of the trailing vortex segments responsible for the induced downwash on the three-dimensional wing.

where the induced downwash w_{ind_j} at each collocation point j is computed by summing the velocity induced by all the trailing vortex segments (see Fig. 12.5). This can be done during the phase of the influence coefficient computations or even later, by using the HSHOE routine with the influence of the bound vortex segment turned off. This procedure can be summarized by the following matrix formulation where all the b_{ij} and the Γ_j are known:

$$\begin{pmatrix} w_{\text{ind}_1} \\ w_{\text{ind}_2} \\ w_{\text{ind}_3} \\ \vdots \\ w_{\text{ind}_N} \end{pmatrix} = \begin{pmatrix} b_{11} & b_{12} & \dots & b_{1N} \\ b_{21} & b_{22} & \dots & b_{2N} \\ b_{31} & b_{32} & \dots & b_{3N} \\ \vdots & & \ddots & \vdots \\ b_{N1} & b_{N2} & \dots & b_{NN} \end{pmatrix} \begin{pmatrix} \Gamma_1 \\ \Gamma_2 \\ \Gamma_3 \\ \vdots \\ \Gamma_N \end{pmatrix}$$

The total lift and drag are then calculated by summing the individual panel contributions:

$$D = \sum_{j=1}^{N} \Delta D_j$$

$$L = \sum_{j=1}^{N} \Delta L_j$$

The induced drag can be calculated, too, by using Eq. (8.146) in the Trefftz plane, which is selected to be far behind the trailing edge and normal to the free stream. Since the wake is force free, the trailing vortex lines will be normal to this plane and their induced velocity can be calculated by using the two-dimensional formula (e.g., Eqs. (3.81) and (3.82)). Consequently, the wake-induced downwash at each of the trailing vortex lines is

$$w_{\text{ind}_j} = \frac{-1}{2\pi} \sum_{i=1}^{N_W} \frac{x_j - x_i}{(z_j - z_i)^2 + (x_j - x_i)^2}$$

where N_W is the number of trailing vortex lines and the influence of a vortex line on itself is set to zero. Once the induced downwash at each of the vortex lines is obtained, the induced drag is evaluated by applying Eq. (8.146):

$$D = -\frac{\rho}{2} \int_{-b_w/2}^{b_w/2} \Gamma(y) w \, dy = -\frac{\rho}{2} \sum_{i=1}^{N_W} \Gamma_j w_{\text{ind}_j} \Delta y_j \qquad (12.10a)$$

If wake rollup routines are used then the results of Eq. (12.10a) may not be unique owing to the nonunique placing of the wake trailing vortices. Therefore, it is recommended that one calculate first the wing circulation with the rolled up wake, and for this induced velocity and drag calculation one should then use the spacing Δy_j of the vortex lines, as released at the trailing edge. (This is the simplest approximation for a force-free wake since many wake rollup routines may not converge to this condition.) Moreover, note that Eq. (12.10a) is similar to Eq. (12.10) but it has a coefficient of $\frac{1}{2}$, which is a result of the first being evaluated at the Trefftz plane (where the trailing vortices seem to be two dimensional) whereas Eq. (12.10) is evaluated at the spanwise bound-vortex line (and there the trailing vortices are observed to be semi-infinite).

This first simple example presented a numerical solution for the lifting-line model, and inclusion of wing sweep and dihedral effects can be done as a homework assignment. Some of the limitations with regard to the wake model and the trailing-edge conditions will be studied in the vortex-ring model that will be presented next. Also, the method presented here does not take advantage of the wing symmetry to reduce computational effort. This important modification is discussed in the following section.

12.2 Modeling of Symmetry and Reflections from Solid Boundaries

In situations when symmetry exists between the left and right halves of the body's surface, or when ground proximity is modeled, a rather simple method can be used to include these features in the numerical scheme. In terms of programming simplicity these modifications will affect only the influence coefficient calculation section of the code.

For example, consider the symmetric wing (left to right), shown in Fig. 12.6, where only the right-hand half of the wing must be modeled. The influence of a panel j at point P can be obtained by any of the influence routines of Chapter 10. For this example, let us use the HSHOE routine of the previous section. Thus the velocity induced at point P by the jth element (with the four corner points A, B, C, and D) is

$$(u_i, v_i, w_i) = \text{HSHOE}(x, y, z, x_{Aj}, y_{Aj}, z_{Aj}, x_{Bj}, y_{Bj}, z_{Bj}, x_{Cj},$$
$$y_{Cj}, z_{Cj}, x_{Dj}, y_{Dj}, z_{Dj}, \Gamma_j)$$

But because of the left/right symmetry, the image panel in the left half wing in Fig. 12.6 will have the same strength, and its effect can be evaluated by calling the influence of the

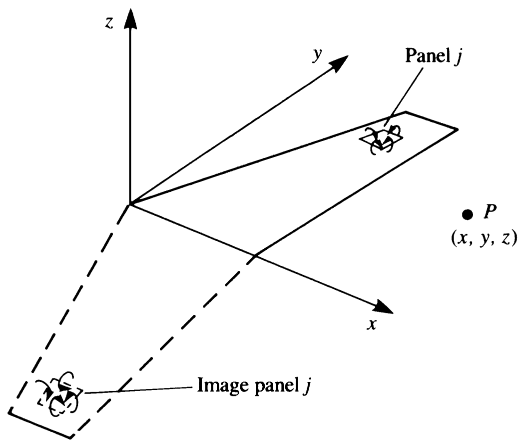

Figure 12.6 Image of the right-hand side of a symmetric wing model.